CuraLoop-RL

ENGLISH USER MANUAL / LOCALHOST POC / v0.3

Screening decisions. Traceable evidence.

Operate the research workbench, follow delayed assay feedback and recover your campaigns with confidence.

LOCAL CPU / HISTORICAL + SYNTHETIC / DURABLE PUBLIC STATE

Independent personal research portfolio. Not an official GPO system or a validated Digital Twin. Prepared 4 November 2026. Screenshots come from the actual English application.

Your path through the workbench

Click a section to jump to it. Use your PDF viewer's bookmarks to return.

03 Start locally

04 Read the workbench

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Start locally

Windows first. One address for the interface and API. No Node.js is needed for the bundled release.

01 Install 64-bit Python 3.12 if needed. The verified release ran on Windows 11 / Python 3.12.10.

02 Open the project folder and double-click **Start-CuraLoop-Web.cmd**. First use installs project-local dependencies.

03 Open **http://127.0.0.1:8000**. Keep the launcher terminal open; Ctrl+C stops the service.

Manual PowerShell alternative

From the source folder:

```
.\Setup-CuraLoop-Web.cmd
```

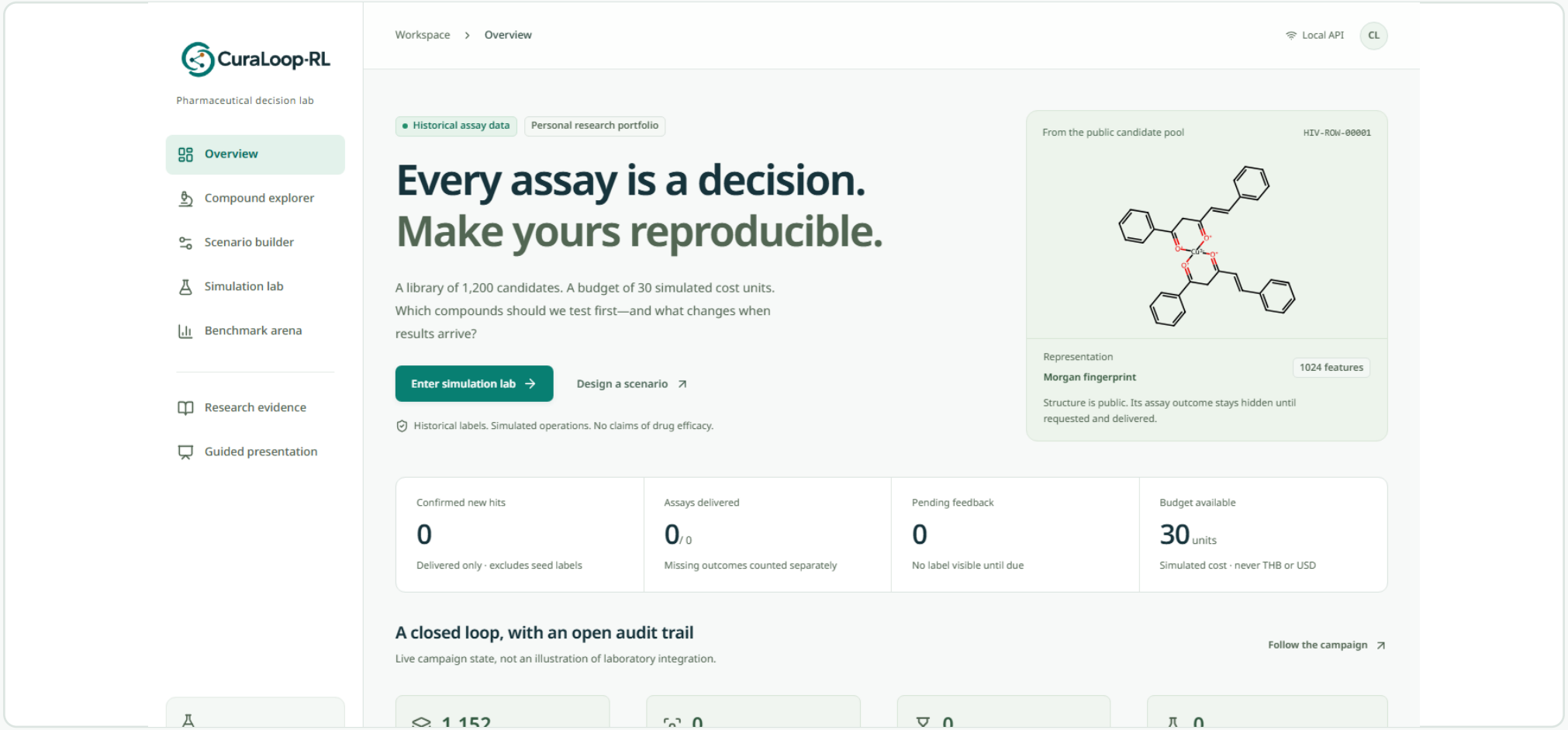
```
.\Start-CuraLoop-Web.cmd
```

Subsequent startup uses local data, models, fonts and built UI assets offline.

The illustrated tutorial uses budget 5 for a compact walkthrough; application defaults use budget 30. Both retain seed 42. Python 3.11 is supported by configuration but was not executed in this verification.

Read the workbench

Begin with the data-origin badge, the active scenario and the live campaign counts.



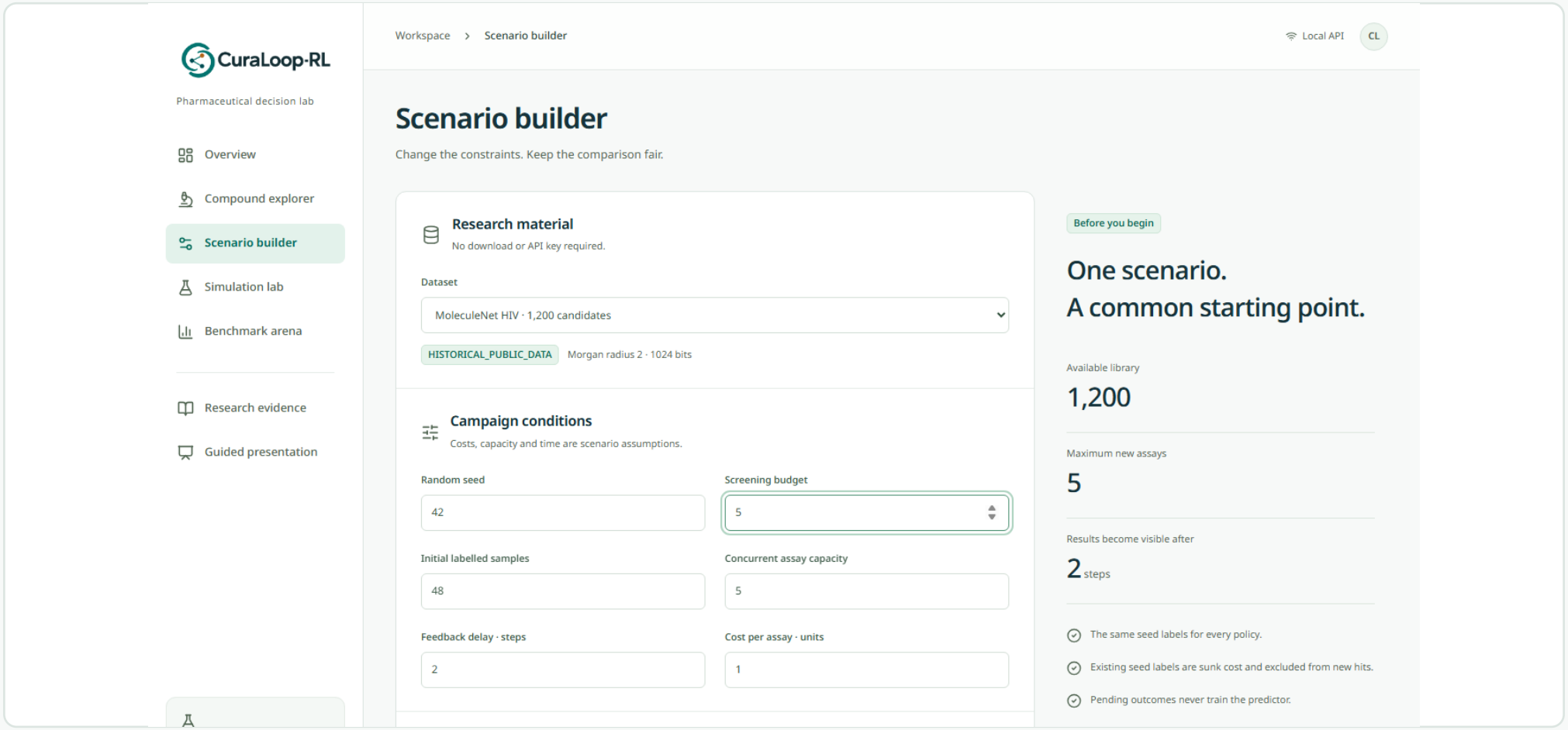
01 Historical assay data and synthetic fixtures have different meanings.

02 Only delivered new active outcomes count as confirmed hits.

03 Enter Simulation lab or design a scenario; both use the same campaign.

Configure a campaign

Use Scenario builder to create a fresh independent campaign. The tutorial keeps seed 42 unchanged.



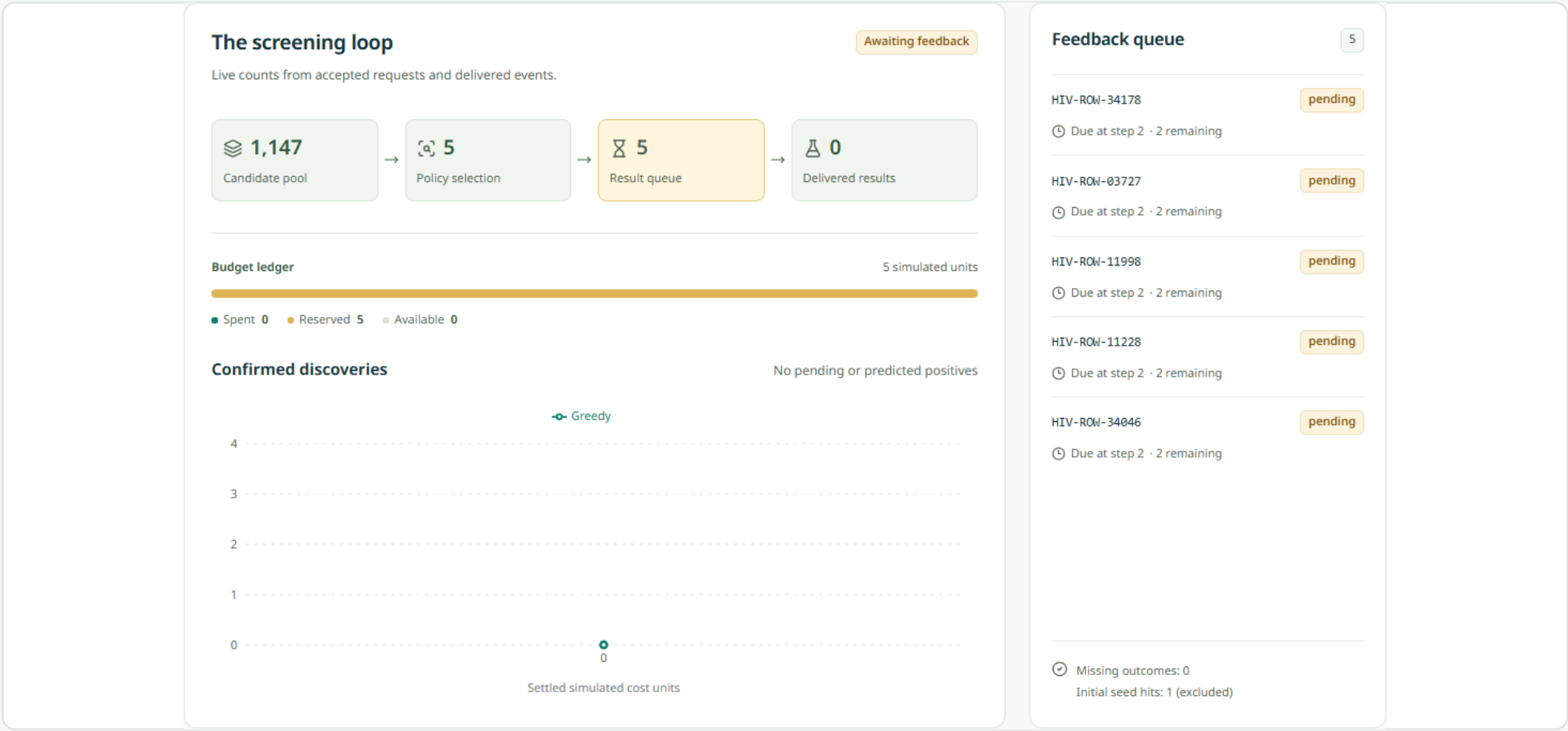
01 Choose HIV or a synthetic dataset and a selection policy.

02 Set budget, queue capacity, feedback delay, jitter and missingness.

03 Scroll down and select Create campaign to apply changes; editing alone does not.

Select and reserve

Select next batch proposes candidates and accepts valid assay requests. Pending labels stay hidden.



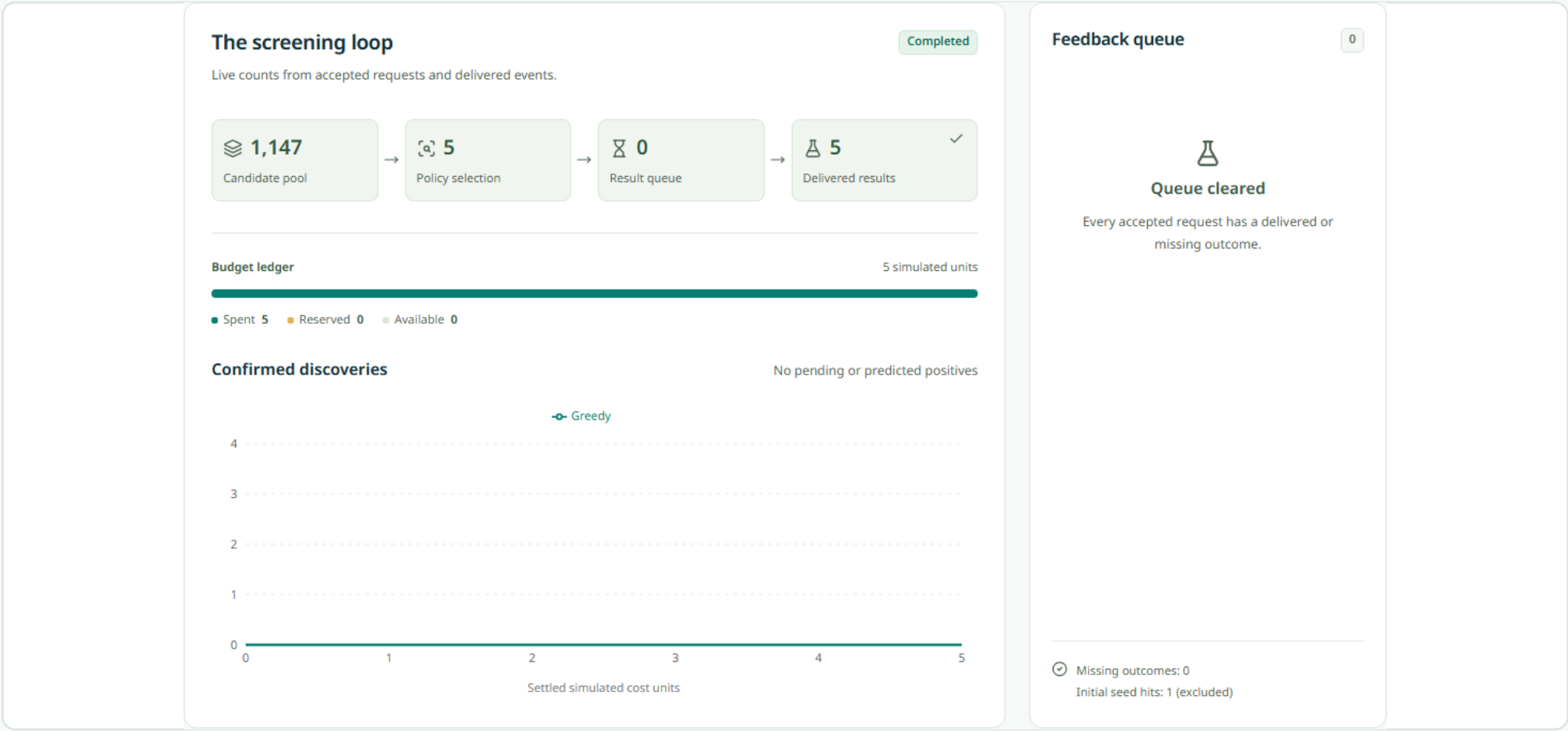
01 With budget 5 and batch capacity 5, this batch reserves all 5 units.

02 Spent stays 0; reserved is 5; available is 0 until feedback settles.

03 Inspect due steps and selection explanations. Predictions are not outcomes.

Advance and receive results

Advance +1 step moves simulated time. With delay 2, two advances deliver the selected batch.



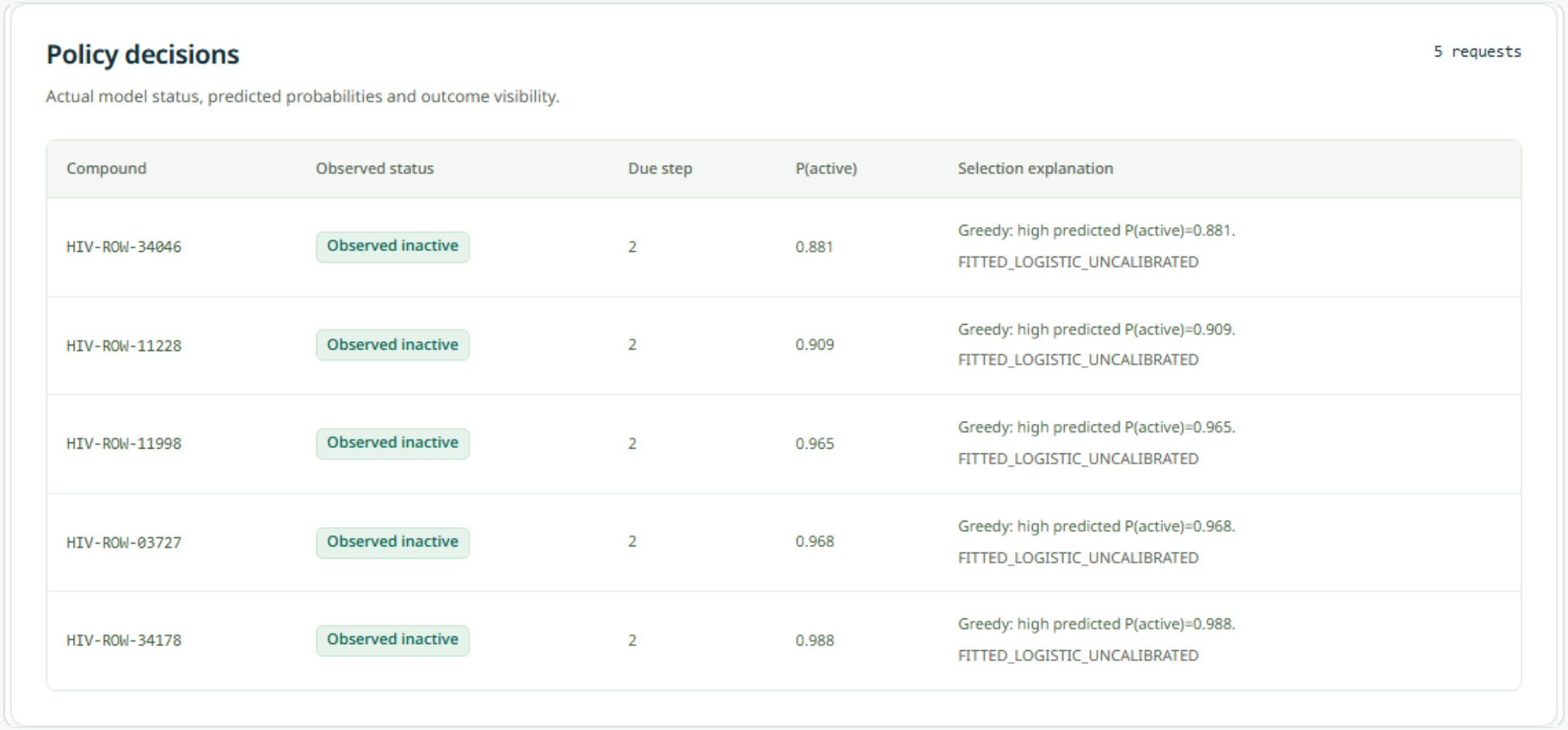
01 The historical example delivered 5 assays and found 0 new active hits.

02 Reserved returns to 0 and spent becomes 5. Budget is never charged twice.

03 Complete run finishes remaining work; Reset creates a fresh state with revisions retained.

Inspect decisions

Read predicted probabilities, observed status and explanations separately. They answer different questions.



Policy decisions

5 requests

Actual model status, predicted probabilities and outcome visibility.

Compound	Observed status	Due step	P(active)	Selection explanation
HIV-ROW-34046	Observed inactive	2	0.881	Greedy: high predicted P(active)=0.881. FITTED_LOGISTIC_UNCALIBRATED
HIV-ROW-11228	Observed inactive	2	0.909	Greedy: high predicted P(active)=0.909. FITTED_LOGISTIC_UNCALIBRATED
HIV-ROW-11998	Observed inactive	2	0.965	Greedy: high predicted P(active)=0.965. FITTED_LOGISTIC_UNCALIBRATED
HIV-ROW-03727	Observed inactive	2	0.968	Greedy: high predicted P(active)=0.968. FITTED_LOGISTIC_UNCALIBRATED
HIV-ROW-34178	Observed inactive	2	0.988	Greedy: high predicted P(active)=0.988. FITTED_LOGISTIC_UNCALIBRATED

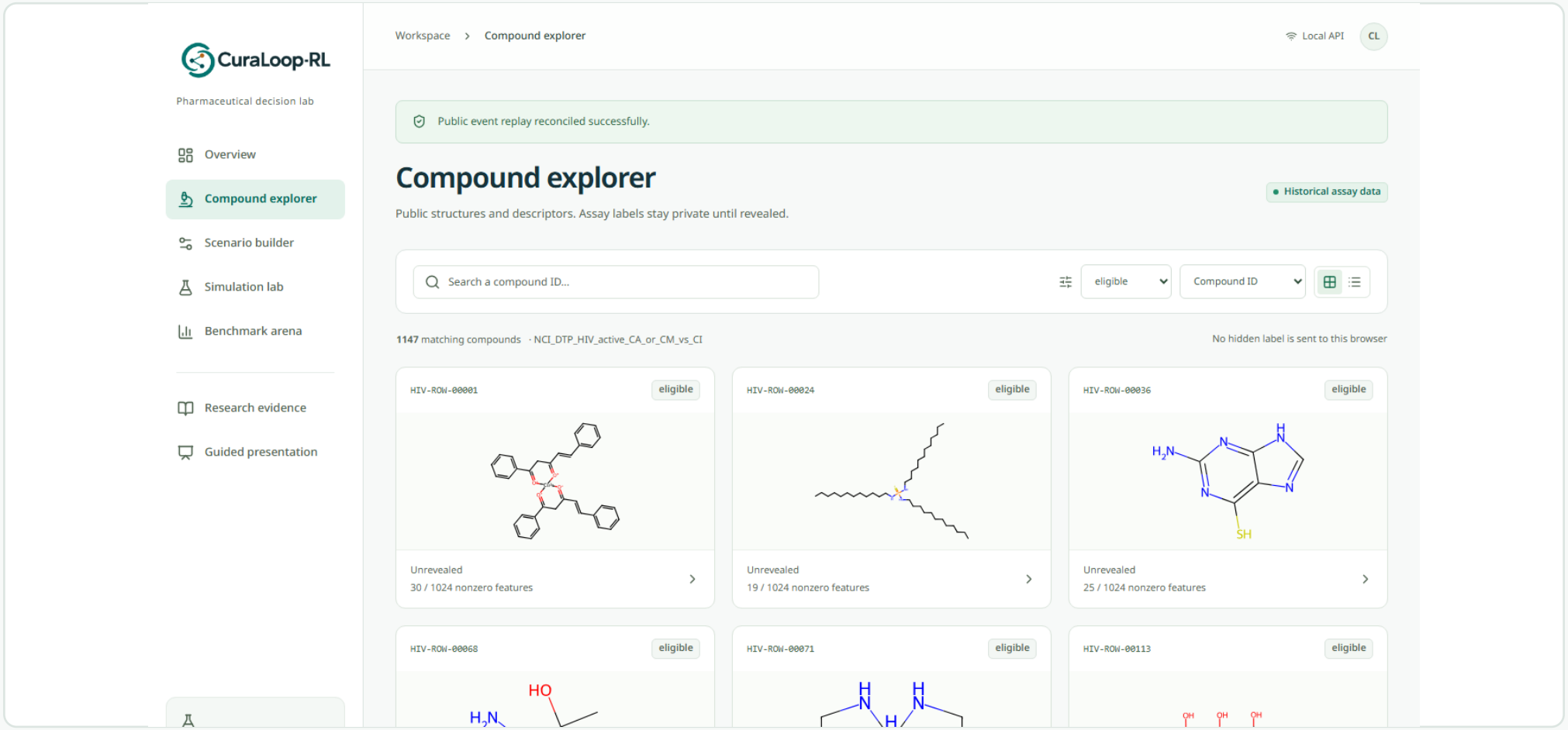
01 Unrevealed, pending, observed active, observed inactive and missing are distinct.

02 Probabilities are uncalibrated model estimates, not confirmed biological activity.

03 Verify replay checks public history; the event timeline records accepted actions and outcomes.

Explore compounds

Compound explorer uses public descriptors and revealed results. Genuine structures come from historical SMILES.



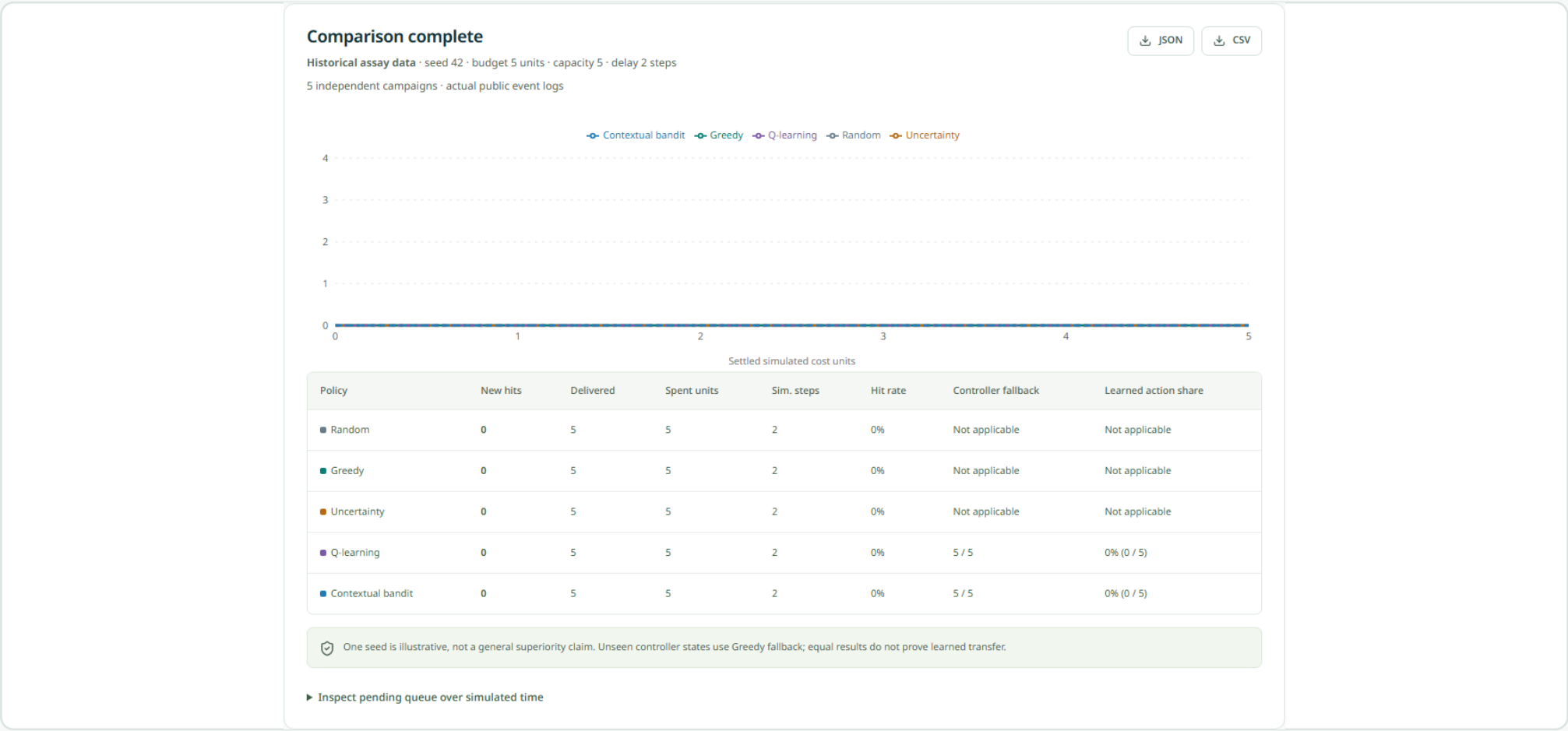
01 Search by compound ID, filter status, choose sorting and move between pages.

02 Inspect opens details; Escape closes the dialog and returns focus.

03 Synthetic fixtures have numerical descriptors and no invented molecular drawings.

Compare independent policies

Benchmark arena executes fresh independent campaigns under matched conditions, including the active scenario seed.



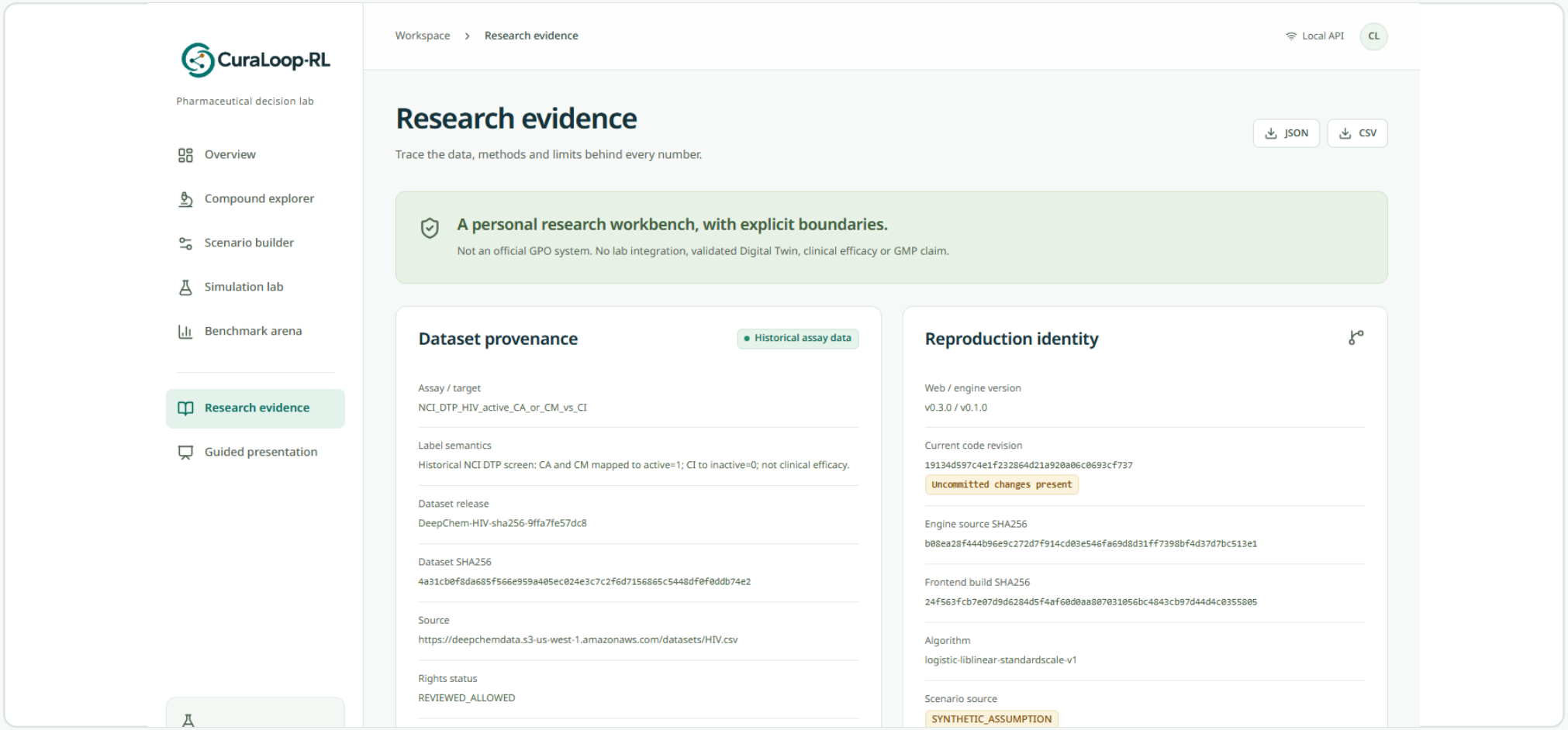
01 Select policies and run the comparison; wait for actual completion.

02 Read new hits, spent units and actual controller fallback / learned-request usage.

03 Saved comparisons survive restart. Recorded HIV charts are separate historical evidence.

Check research evidence

Research evidence exposes provenance, implementation identity, model limitations and recovery tools.



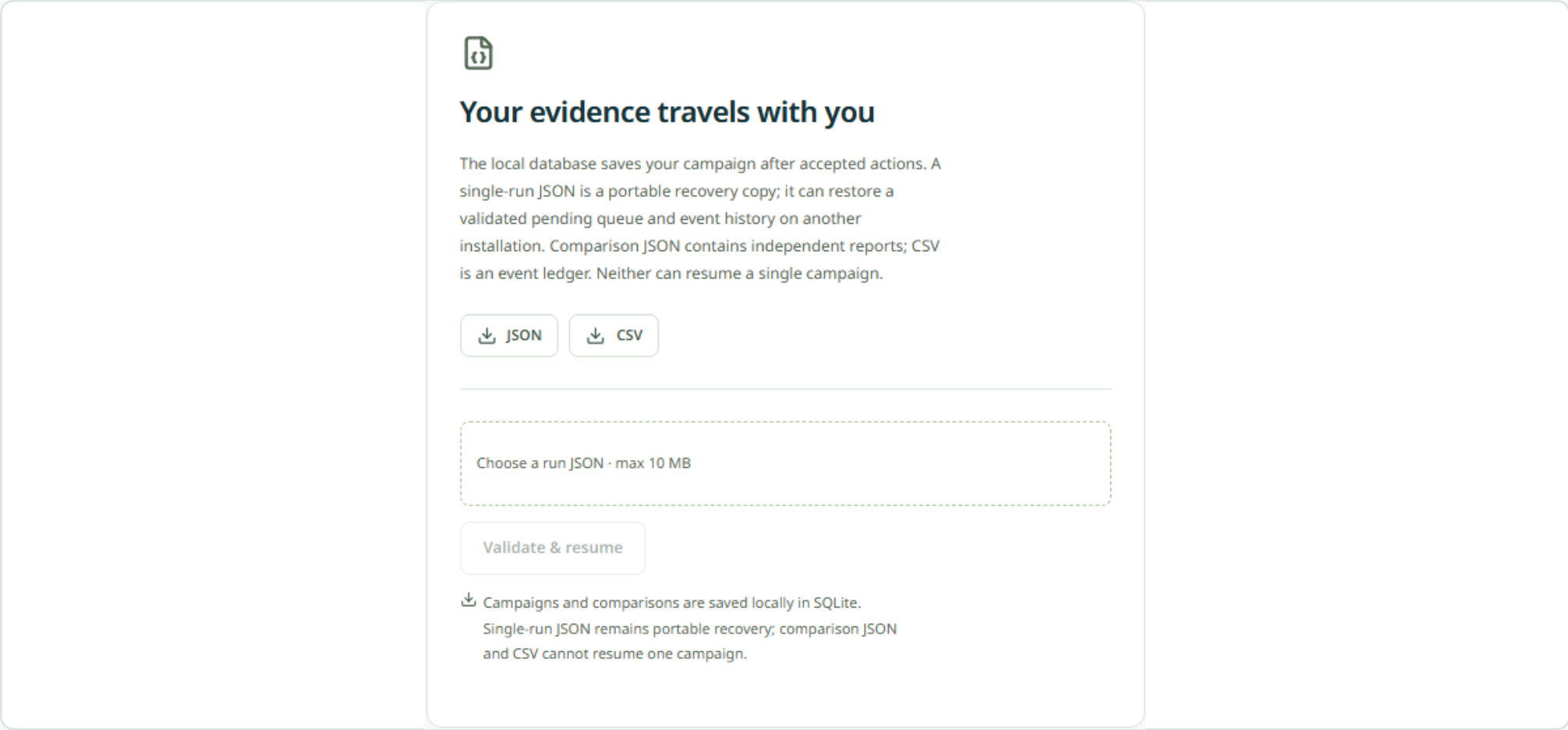
01 Check dataset release, hashes, label semantics and seed-selection rules.

02 Review weak held-out performance and synthetic-only controller training.

03 Use the cited primary sources; current simulation output is not clinical validation.

Export and resume JSON

A single-run JSON is portable recovery. Comparison JSON and CSV have different purposes.



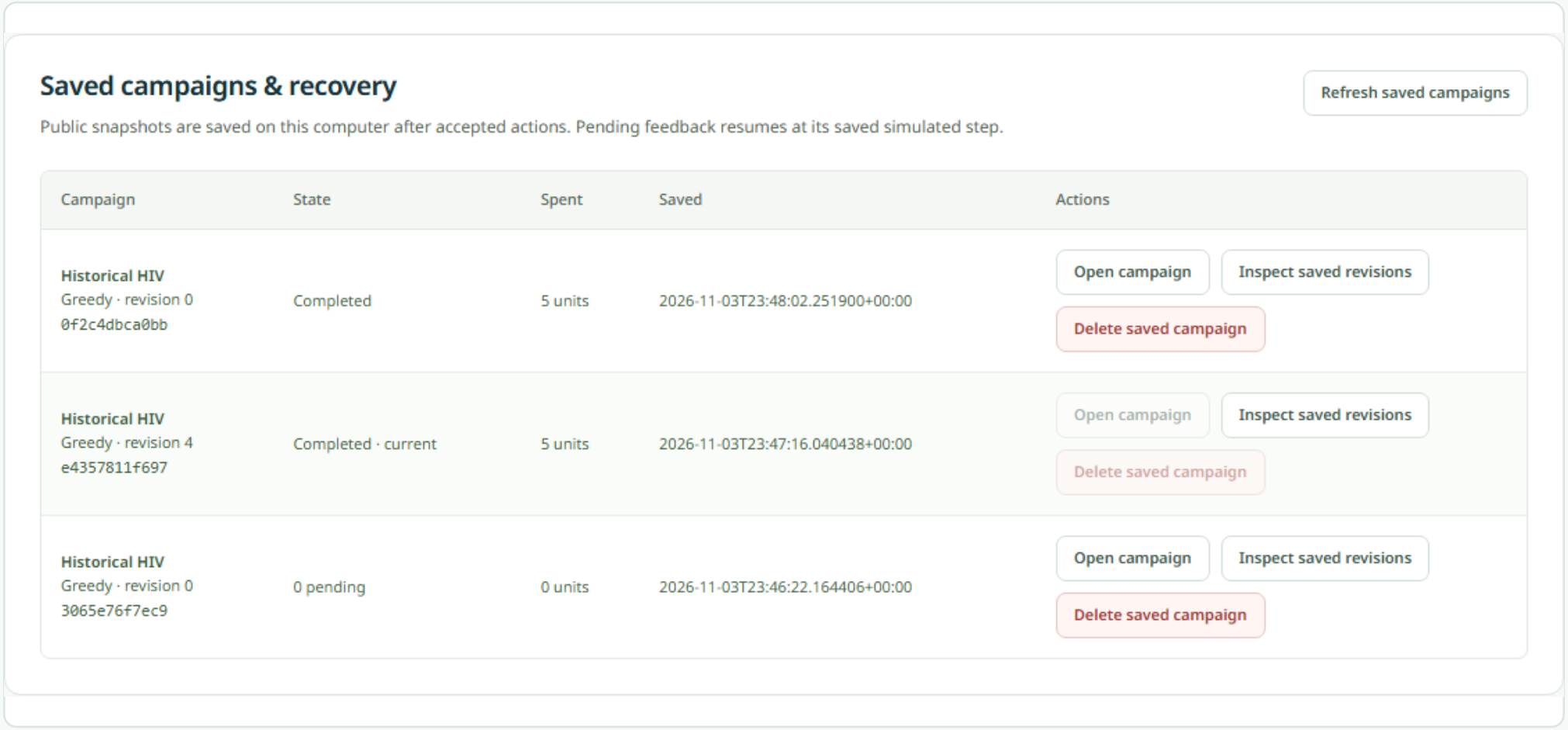
01 Download JSON for the current campaign; CSV is its public event ledger.

02 Choose the single-run JSON and select Validate & resume.

03 Validation reconstructs pending requests and event history; invalid imports preserve the active run.

Open saved campaigns

The primary web app saves accepted actions automatically in local SQLite. No cloud account is required.



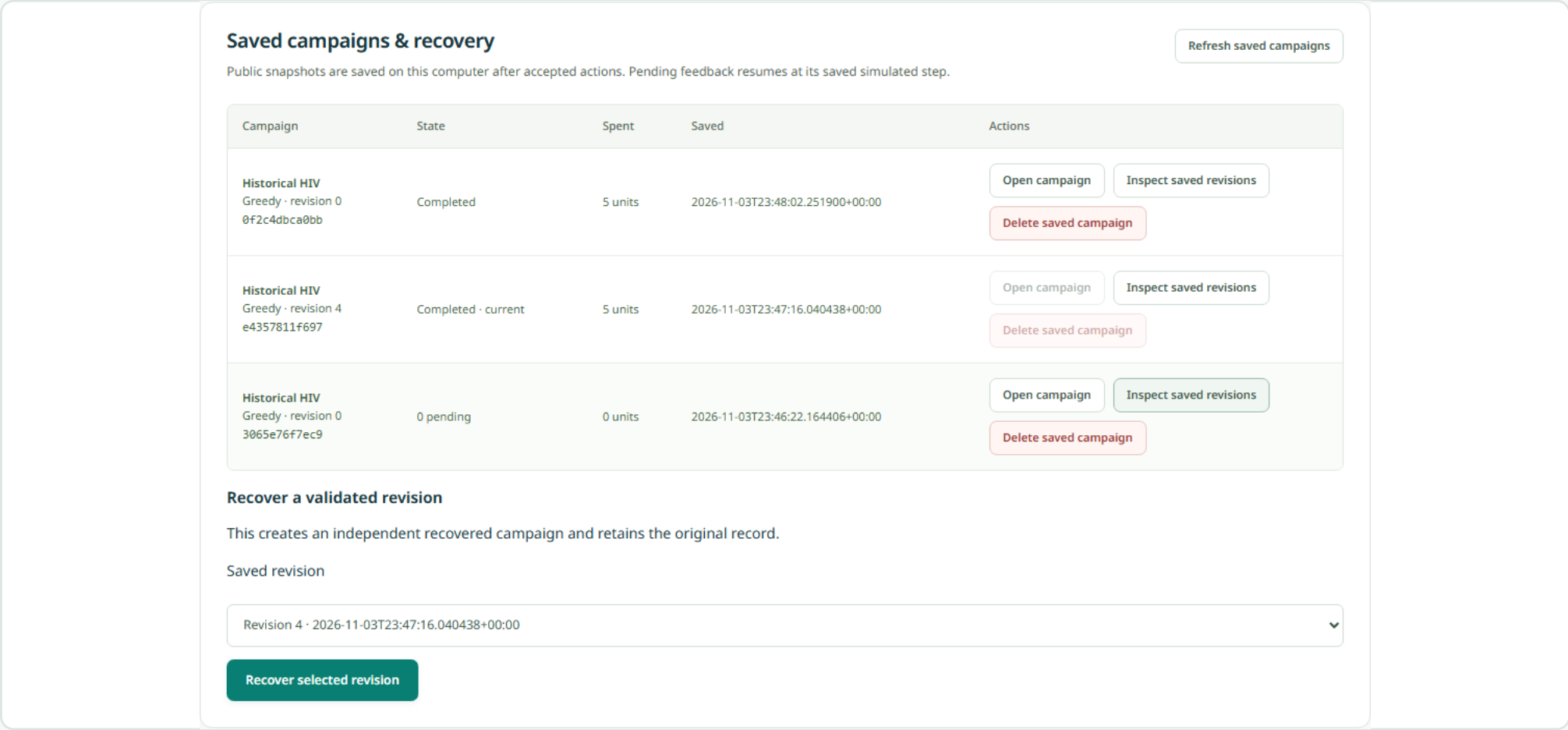
01 Use Saved campaigns & recovery to open a previous campaign.

02 Refresh saved campaigns updates the list; current campaigns are labelled.

03 Export before explicit deletion. Deletion permanently removes the saved campaign and revisions.

Recover a saved revision

Inspect saved revisions lets you recover earlier validated state, including after Reset or a damaged current row.



01 Choose a campaign, then select a saved revision.

02 Recover selected revision creates a new independent campaign.

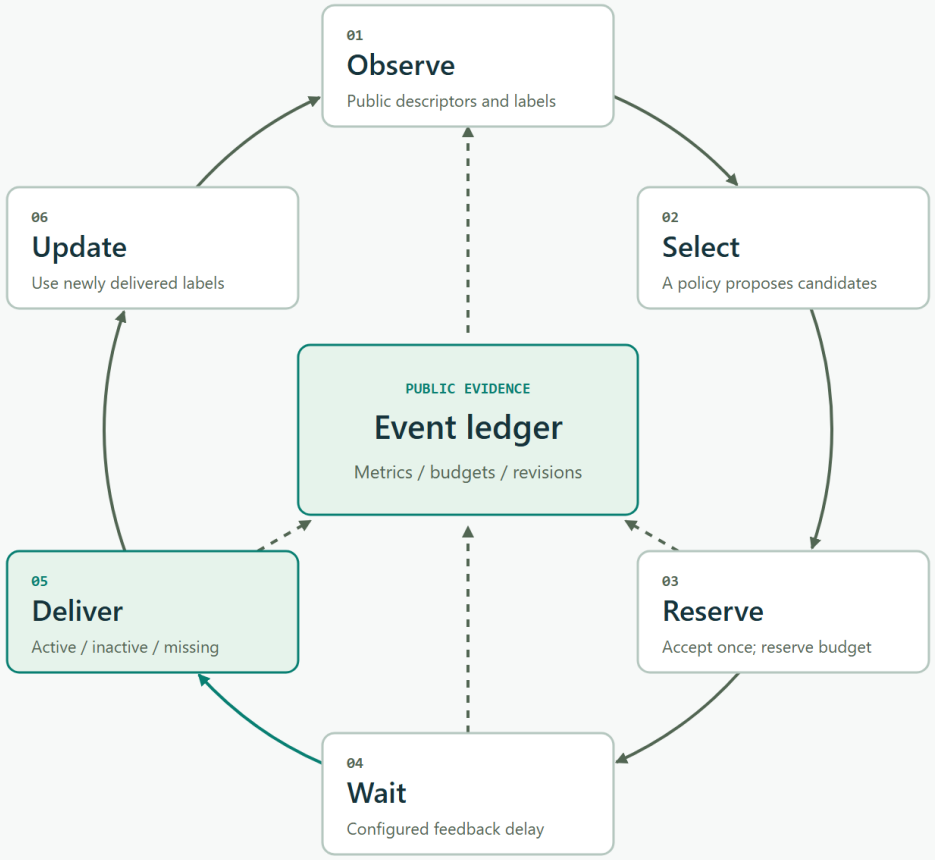
03 The original record remains intact. Recovery and active campaign actions are serialized.

Understand the screening loop

CURALOOP-RL / RESEARCH WORKBENCH

From candidates to traceable evidence

An auditable screening cycle with delayed feedback and bounded resources.



Solid: next step. Dashed: record accepted actions and due results; read public state.

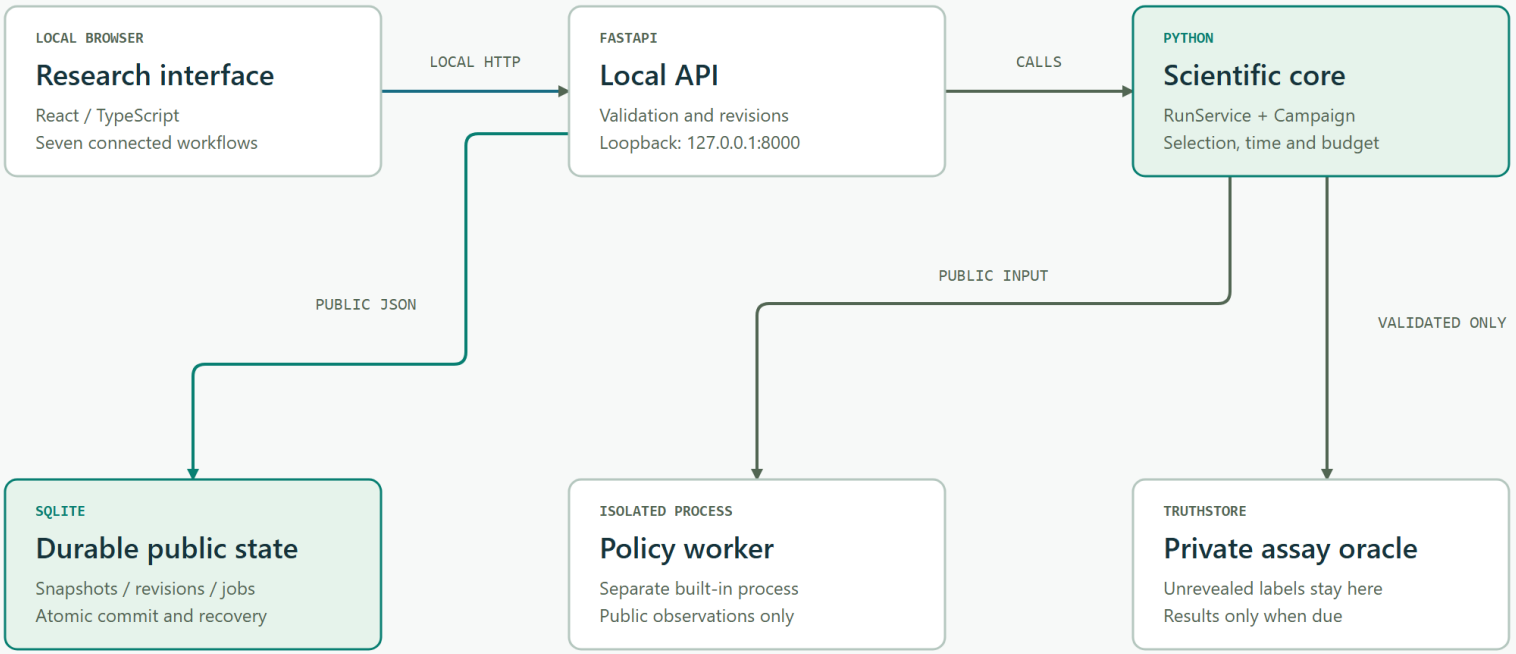
Repeat while budget and candidates permit. Pending predictions and seed hits are not new confirmed hits.

Understand the architecture

CURALOOP-RL / RESEARCH WORKBENCH

A local workbench, one scientific core

One local address. One Python authority for scientific state.



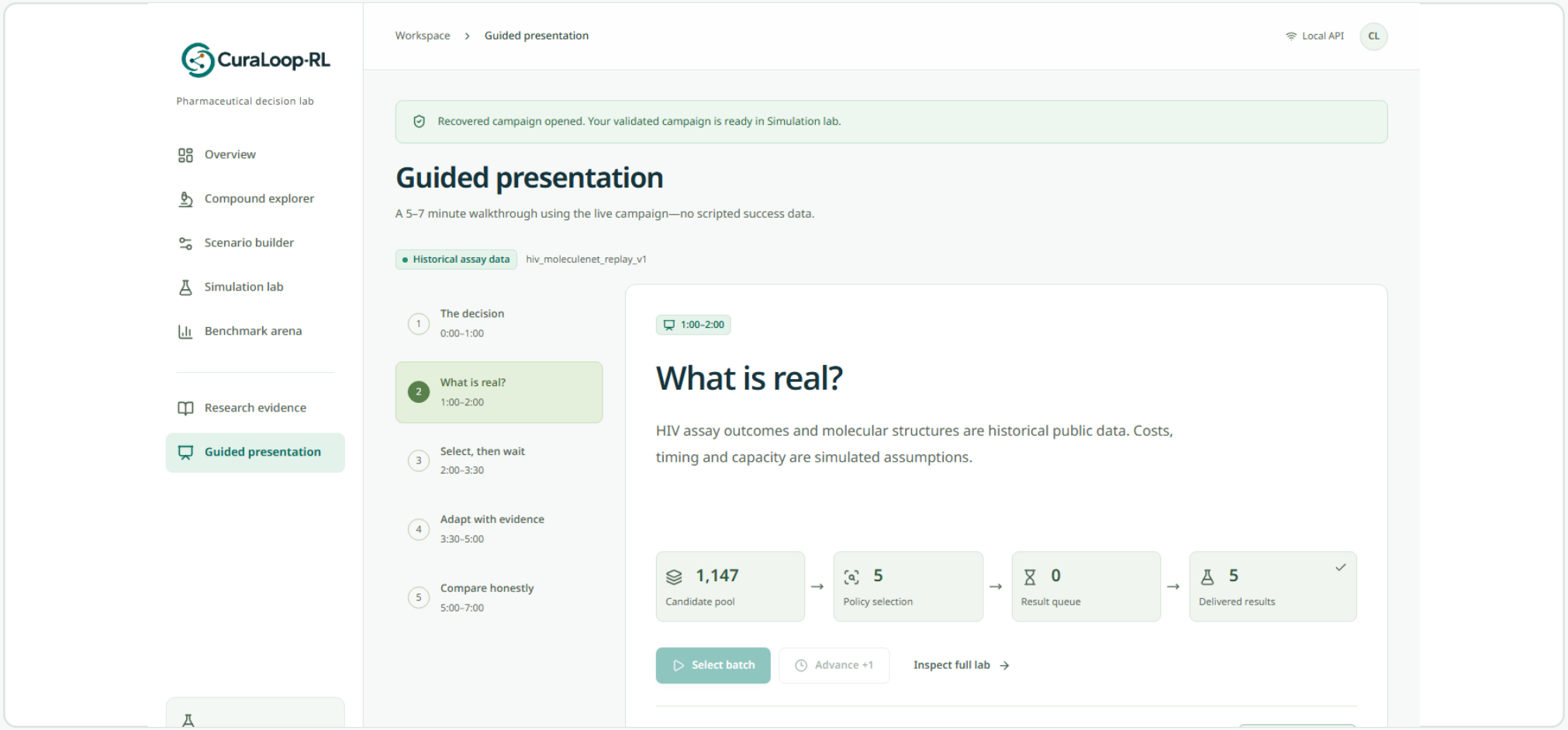
Only seed labels and delivered outcomes can inform policy decisions.

Arrows show permitted calls. Responses return through the scientific core and API.

Local process separation is an engineering boundary, not an OS sandbox for hostile code.

Present the project

Guided presentation uses the live campaign. It provides a 5-7 minute story rather than scripted success metrics.



01 Explain the decision and distinguish historical labels from simulated operations.

02 Demonstrate selection, delayed feedback and the public audit trail.

03 Finish with matched comparisons, exports and the recorded controller fallback limitation.

Interpret historical and synthetic data

A source label changes what a result means. Simulated operational settings never become measured laboratory evidence.

01 Historical HIV: real retrospective assay labels and genuine molecular structures; simulated budget, capacity and timing.

02 Synthetic fixture: fictional descriptors and labels for engineering checks. No biological assay or molecular-structure claim.

03 Missing outcome: settles cost, but is neither an inactive label nor an active hit. It is excluded from training labels.

04 Model limits: uncalibrated probabilities and heuristic uncertainty. Recorded learned controllers used Greedy fallback on 300/300 historical requests each.

Actual synthetic verification examples: observed fixture: 5 new hits / 18 delivered; missing fixture: 18 missing / 0 hits. Both spent 18 simulated units. These are labelled fixtures, not new biological benchmarks.

Back up and troubleshoot

Keep portable recovery copies and preserve damaged data. The primary web API uses one active server and one worker.

01 Portable backup: export single-run JSON. Comparison JSON has multiple reports; CSV is an event ledger. Neither resumes one campaign.

02 Database backup: stop the server cleanly, then copy `.local` with its database and any sidecar files.

03 Interrupted comparison: the saved job is marked interrupted, never silently rerun. Start a new comparison explicitly.

04 Corrupt database: stop and copy it before investigation. Point CURALOOP_STORAGE_PATH to a new file and validate a portable JSON; keep the original.

05 Startup issues: retry Setup after a download failure. Reconnect if the server is unavailable. A port conflict never authorizes killing another application.

Windows operator guide | Persistence and recovery reference

Verification and reference

The manual documents software operation. Its screenshots and verification are separate from frozen research benchmark evidence.

01 Scientific scope: retrospective research simulator. No clinical efficacy, GMP suitability, real savings or operational Digital Twin validation.

02 Capture provenance: actual Windows Chromium, 1440x1000 desktop and 390x844 mobile. New branding screenshots use isolated local test storage.

03 Independent runs: comparison policies share conditions but never share new outcomes. The example historical campaign retains its zero-hit result.

04 Accessibility and limits: keyboard/dialog/overflow and automated accessibility checks are exercised. Manual screenreader and physical-device checks remain unrun.

Installation and troubleshooting

Evaluation and limitations

Research sources and dataset provenance

Release verification

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